

Real-time ultrasound-guided hemostasis using suture-mediated closure device

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Abstract

Purpose: Suture-mediated hemostasis device takes a long time to train and have limitations in finding appropriate compression point depending on the tactile sense of the hands. If the appropriate compression point is determined using ultrasound and the instrument is used under ultrasound guidance, it is expected to reduce the technical failure and additional manual compression.

Materials and methods: This retrospective study included 104 patients in whom the ProGlide vascular closure device (Abbott Vascular, Redwood City, CA, USA) was deployed to close common femoral artery access between January 2022 and June 2022. Ultrasound-guided hemostasis was performed in 54 patients and 50 conventional hemostasis was performed. We analyzed the medical records (procedure, patient body mass index, coagulation function, visual imaging, ultrasound imaging, time to achieve hemostasis) and post-treatment medical records (progress records, nursing records) to investigate the technical success rate and complications (hematoma, pseudoaneurysm formation, SFA occlusion, access site infection) of the procedure.

Results: Technical success rate was 52/54 (Ultrasound guided group) and 47/50 (conventional group), respectively ($p > 0.05$). Mean hemostasis time was 271 s (US guided group) and 317 s (conventional group), respectively ($p > 0.05$). Additional manual compression was applied in 7/54 (US guided group) and 15/50 (conventional group) ($p < 0.05$). In both groups, complications such as hematoma formation, access site infection and SFA occlusion did not occur.

Conclusion: Using ultrasound guidance when using the suture mediated closing device reduces the frequency of additional manual compression without increased risk of complication.

Keywords

Techniques and procedures, interventional radiology, closure device, arterial access, ultrasound guidance

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Introduction

Most endovascular procedures, ranging from simple diagnostic angiography to endovascular aneurysm repair use the common femoral artery as an access route. However, femoral artery access site bleeding complications carry increased morbidity and mortality compared to those with other access routes.¹ Various sizes of vascular sheaths and devices are inserted depending on the procedure, and the entire process is successfully terminated by safe hemostasis after the procedure. Therefore, it is important to achieve safe and efficient hemostasis without complications such as bleeding, hematoma, or pseudoaneurysm formation.

Since the introduction of percutaneous transarterial access for endovascular procedures, manual compression has become the gold standard for achieving hemostasis.

In addition to the traditional manual compression method, various hemostatic devices have been used. Vascular closure devices have the advantages of better patient comfort, shorter time to achieve hemostasis, and earlier ambulation and hospital discharge. In particular, suture-mediated closure (SMC) devices are widely used because they can safely and effectively achieve hemostasis even when a very large-diameter vascular sheath is inserted.^{2,3}

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Table 1. Patient demographics.

	US-guided group	Conventional group
Sex		
Male	39	36
Female	15	14
Age (mean, range), years	68.28 (24–87)	67.44 (35–85)
BMI (mean, range), kg/m ²	24.26 (18.09–33.03)	24.31 (16.21–32.76)
Procedure		
Elective TACE	34	31
Emergency bleeding embolization	15	17
Angiography	0	2
Percutaneous angioplasty	4	0
Covered stent insertion	1	0
Coagulopathy		
Thrombocytopenia (<50,000/dL)	3	4
INR > 1.5	4	3
Both thrombocytopenia and INR > 1.5	1	1
Medication		
Anticoagulation	2	1
Antiplatelet	1	2
Anticoagulation and antiplatelet	4	2
Diabetes	22	14

INR: international normalized ratio.

However, SMC takes a long time to reach the plateau of the learning curve⁴ and has limitations in finding appropriate suture knot advancement points, depending on the tactile sense of the hands during the operator's early experience.

Purpose

If the appropriate suture knot advancement point is determined using ultrasound and performed under ultrasound guidance, it is expected to shorten the learning curve and reduce additional manual compression and hemostasis time. The purpose of this study is to analyze the efficacy of using SMC under ultrasound guidance.

Materials and methods

Patients

At our institution, 247 femoral artery accesses were performed between January 2022 and June 2022. This retrospective study included 104 patients (75 men and 29 women), with a mean age of 67.8 years, in whom the Perclose ProGlide vascular closure device (Abbott Vascular, Redwood City, CA, USA) was deployed to close the common femoral artery access during that period (Table 1). US-guided hemostasis was performed in 54 patients; conventional non-US-guided hemostasis was used in 50 patients. Vascular access was used for elective transarterial chemoembolization (TACE, $n=65$), emergency bleeding

embolization ($n=32$), percutaneous angioplasty ($n=4$), diagnostic angiography ($n=2$), and covered stent insertion ($n=1$) (average sheath size 5.06 F, range 5F–8F). Coagulopathy was observed in 16 patients (seven thrombocytopenia, seven International normalized ratio prolongation, two both of all). Twelve patients were administered antiplatelet or anticoagulation drugs.

A suture-mediated device was not used in patients with a common femoral artery (CFA) diameter of <6 mm, calcification on the anterior surface of the CFA, unstable vitality, use of 4F size sheath, or patient with irritability.

This study was approved by the Institutional Review Board (2022AN0300) and waived for informed consent.

Technique

Two experienced interventional radiologists performed femoral artery access and closure device deployments. All procedures were performed under local anesthesia. At the beginning of the procedure, a common femoral artery puncture (demarcated by the inferior two-thirds of the femoral head and superior to the bifurcation) was performed under ultrasound guidance (Figure 1). If lymph nodes were observed in the inguinal area on ultrasound, care was taken not to allow the puncture pathway to pass through them. After the procedure, a 0.035-inch soft guiding wire (Terumo, Tokyo, Japan) was inserted into the vascular sheath. The sheath was removed along the wire, and an SMC device was inserted into the common femoral artery along the wire. The deployment of the suture knot

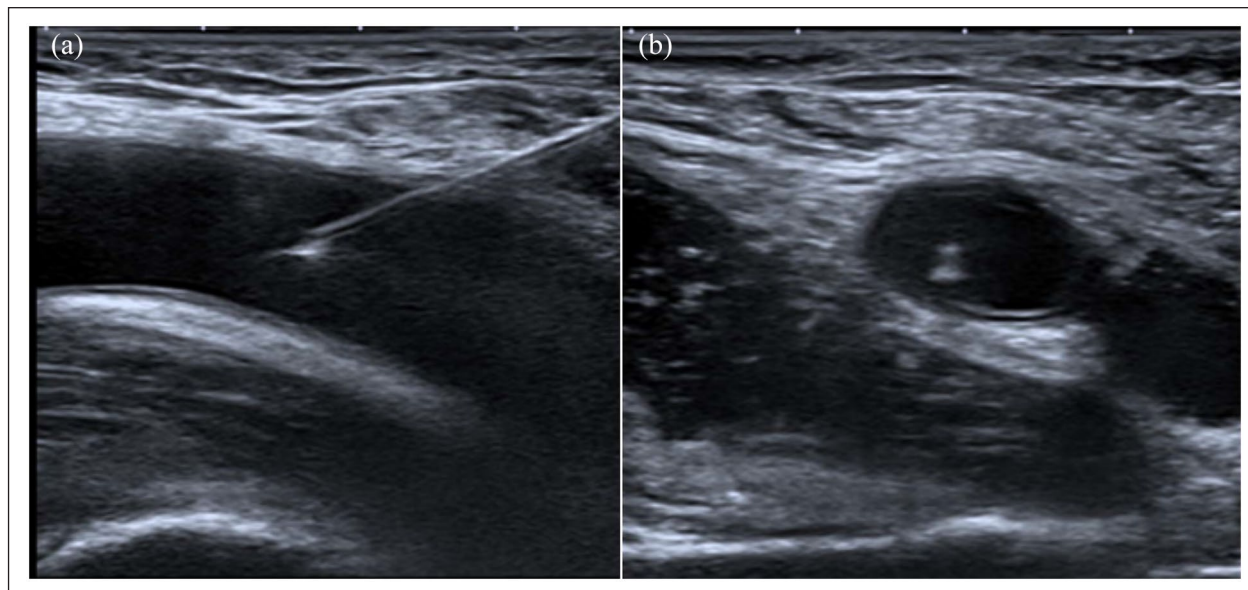


Figure 1. (a) Axial and (b) longitudinal view of US-guided common femoral artery puncture.

In the axial view, it can be seen that the 12 o'clock direction of the common femoral artery is accurately punctured, and in the longitudinal view, it can be seen that there are no obstacles such as lymph nodes in the access route.

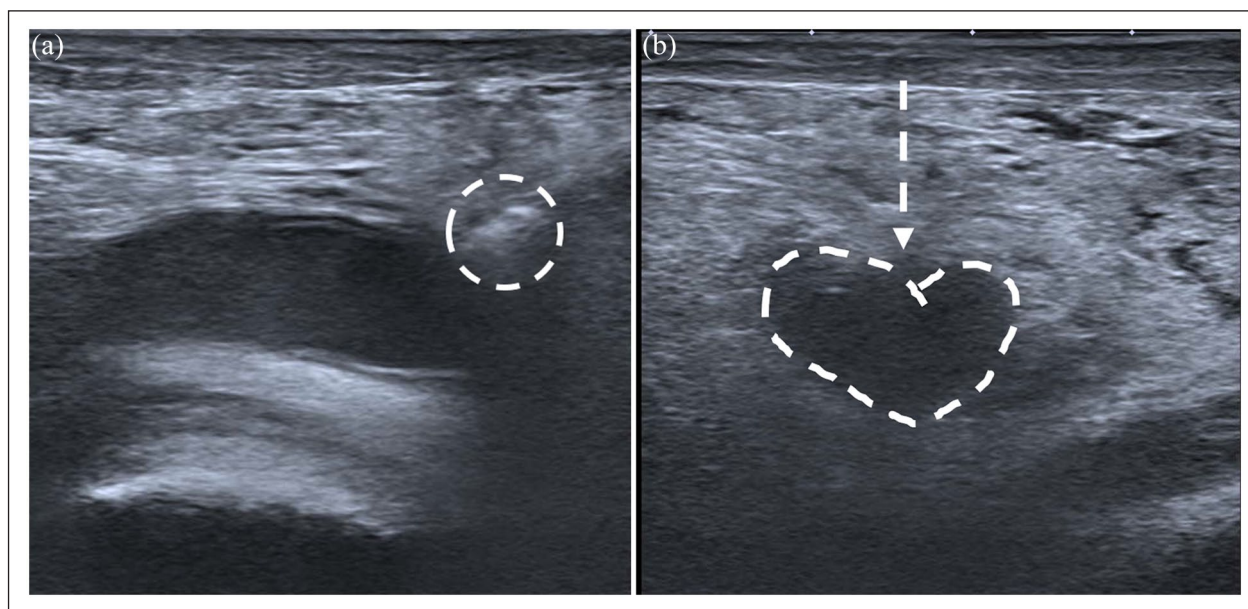


Figure 2. (a) Axial and (b) longitudinal view when using the "US-guided knot advancement technique."

In the axial view, it can be seen that the 12 o'clock direction of the common femoral artery is accurately pressed by the knot trimmer and changed to a "heart shape" (dashed heart). In the longitudinal view, it can be seen that the tip of knot trimmer (dashed circle) accurately advances to the anterior wall of common femoral artery.

was implemented using an existing method. In the ultrasound-guided group, we gently advanced the suture knot trimmer until its tip reached the anterior wall of the common femoral artery under ultrasound guidance (Figure 2). In the conventional group, the trimmer was advanced until its tip was felt manually against the wall of the common femoral artery. After completing the proper suture knot

advancement using a trimmer in each group, the suture knot was cut. Additional manual compression was applied in cases where there was continuous bleeding through the skin opening, even after all blood collected in the puncture tract was removed with gauze during the hemostatic process, regardless of the bleeding characteristics (e.g. bleeding, oozing, pumping).

Table 2. Binary logistic regression for additional manual compression.

	B	S.E	Wals	DF	p	Exp (B)
Performing doctor	1.088	0.639	2.901	1	0.089	2.967
Sex (female)	1.610	0.642	6.294	1	0.012*	5.005
Age	-0.003	0.026	0.015	1	0.904	0.997
US apply (conventional)	1.460	0.646	5.104	1	0.024**	4.305
Sheath size	-19.295	11,121.556	0	1	0.999	0
Diabetes	-0.052	0.707	0.005	1	0.941	0.949
INR prolongation	-1.104	1.072	1.059	1	0.303	0.332
Lymph node in access route	-0.544	0.644	0.712	1	0.399	0.581
Antiplatelet	1.596	1.243	1.648	1	0.199	4.933
Anticoagulant	-1.870	1.296	2.082	1	0.149	0.154
Thrombocytopenia	-1.028	0.941	1.194	1	0.275	0.358
BMI	-0.242	0.139	3.022	1	0.082	0.785
Skin-puncture site distance	1.255	0.956	1.723	1	0.189	3.507

*Additional manual compression was statistically significantly more frequent in female patients.

**Additional manual compression was statistically significantly more frequent in conventional techniques.

Additional manual compression was performed until the bleeding stopped completely. Immediately after the hemostatic process was completed, ultrasonography was used to evaluate immediate complications such as hematoma and superficial femoral artery occlusion. Device use time and total hemostasis time were recorded.

Definitions

Technical success was defined as the achievement of complete hemostasis without additional manual compression or with additional manual compression for less than 5 min.

Clinical success was defined as the achievement of complete hemostasis without complications regardless of additional manual compression.

Device usage time was defined as the time between vascular sheath removal and suture knot cutting.

The total hemostasis time was defined as the time from vascular sheath removal to achieving complete hemostasis.

The skin-puncture site distance was defined as a straight line denoting the distance from the site of skin puncture to the site of vessel puncture, as measured by the ultrasound sagittal view.

Outcome

We retrospectively analyzed the medical records (type of procedure, body mass index (BMI), coagulation function, ultrasound image, time to achieve hemostasis, progress records, and nursing records) to investigate the technical and clinical success rates, hemostasis procedure efficacy (additional manual compression and total hemostasis time), and complications (hematoma, pseudoaneurysm formation, SFA occlusion, and access site infection) of each procedure.

Statistical analysis

Pearson's chi-square test was used to compare technical success rates between the two groups.

The Mann-Whitney *U* test was used to analyze the total hemostasis time difference between the two groups, and binary logistic regression was used to analyze the factors affecting hemostasis procedure efficacy using SPSS (version 20) (SPSS, Inc., Chicago, Illinois).

Results

All 104 hemostatic procedures in both groups were clinically successful, and complications such as hematoma, pseudoaneurysm formation, and infection were not reported in the follow-up clinical records. There was no significant difference in clinical success rates between the two groups.

The technical success rates were 96.7% (52/54, ultrasound-guided group) and 94% (47/50, conventional group), respectively ($p > 0.05$). The mean total hemostasis time was significantly shorter in the ultrasound-guided group (255.9, range: 90–2629 s) than that in the conventional group (317.2, range: 105–1748 s) ($p < 0.05$). Additional manual compression was applied significantly more frequently in the conventional group (15/50) than in the US-guided group (7/54) ($p < 0.05$). In the binary logistic regression, statistically significant additional manual compression was performed in women and in those who did not receive US guidance (Table 2). Age, BMI, skin-puncture distance, coagulopathy, medication status, and performing physician did not have a statistically significant effect on additional manual compression.

Discussion

Manual compression is the gold standard for achieving hemostasis. Although this method successfully achieves

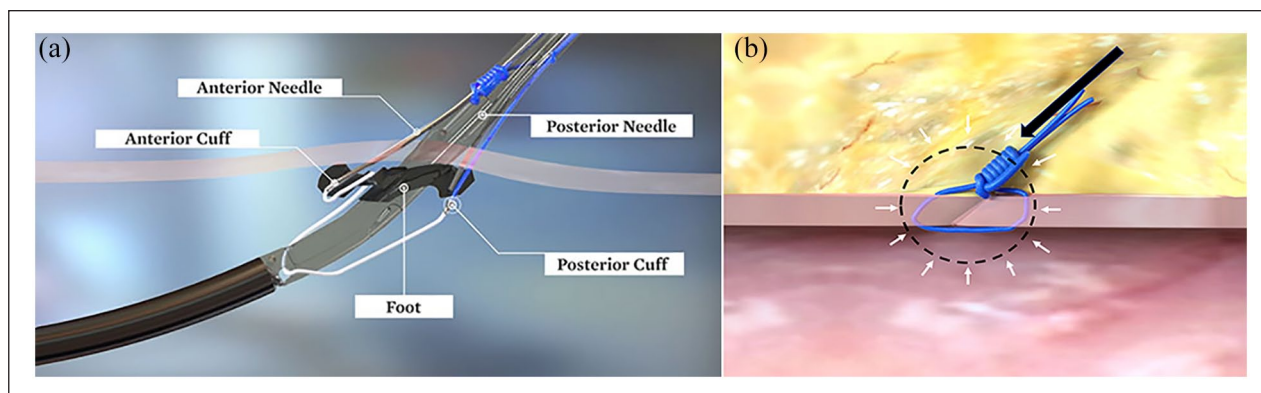


Figure 3. (a) When the plunger is fired, a suture knot is installed in the puncture area by the anterior and posterior needles and (b) by advancing the suture trimmer (black arrow), deployed suture knots are tightened across both sides of the artery wall (black washed circle). The closer the trimmer tip is to the area where the arterial wall is interrupted by the sheath insertion, the more approximate the arterial walls on both sides (white arrows) and the effect of SMC is maximized.

hemostasis in most cases, it has limitations. First, manual compression requires sustained partially occlusive pressure over the arterial access site for approximately 15–20 min, followed by 4–6 h of patient immobilization.⁵ Pressure on the patient's groin and the long absolute bed rest increase the patient's discomfort and prolong hospital stay. In addition, patient noncompliance can cause rebleeding of the access site (from small subcutaneous hematoma formation to life-threatening bleeding). Further, if the patient has a coagulopathy or is administered anticoagulation or antiplatelet agents, the complication rate increases when manual compression is performed.⁶ Third, skilled personnel must stand by the patient for tens of minutes for safe and successful hemostasis, and it is essential to monitor the patient even after hemostasis. To overcome these limitations, various vascular closure devices have been developed and are now widely used.

The Perclose ProGlide device was the first SMC approved by the US Food and Drug Administration in 1997. A pre-tied knot made of monofilament polypropylene was installed in the defect of the arterial wall caused by the arterial access, and the wall was approximated by advancing the knot close to the access site to induce hemostasis (Figure 3). However, each step should be performed carefully, and there is a possibility of technical failure when used by an inexperienced person. Balzer et al.⁴ reported a “learning curve phenomenon” with SMC. Based on the mechanism of operation of Perclose ProGlide, we devised the “US-guided knot advancement technique” to move suture knots as close and accurately as possible to the real arterial access site. By using ultrasound image guidance to advance the tip of the trimmer to the arterial access site in real time, less additional manual compression and shorter hemostasis time could be obtained than when only the tactile sense of the hand was used for hemostasis. This seems to be due to the accurate approximation of the arterial wall using SMC under ultrasound guidance. The exact location of the trimmer tip can be

determined using our technique, and the effectiveness of SMC can be maximized by advancing the trimmer tip accurately to the vessel wall of the puncture site. In our institution's experience, experts with sufficient experience in ultrasound guidance needed less than five cases to reach the plateau of the learning curve of “US-guided knot advancement technique.”

In our study, additional manual compression to achieve complete hemostasis was performed more frequently in female patients. Similar to our findings, previous studies and Instructions for Use of Perclose ProGlide reported increased vascular complications, higher vascular closure device failure, and longer hemostasis times in females.^{6,7}

Limitations

In previous studies, obesity was reported as a risk factor for vascular closure failure and Perclose ProGlide-related complication.^{7–9} In our study, only 6 of 104 patients had BMI > 30 kg/m²; thus, the effect of BMI may not have been accurately reflected.

Most of the subjects were 5F sheaths because, at our institution, we always use our technique to achieve hemostasis to large-bore sheath (>7F) access sites. Therefore, the effect of sheath size on additional manual compression and hemostasis time appears to have been underestimated. Further prospective research is needed to prove the effect of this technique on the hemostasis of large-bore sheaths.

Further limitations include the small sample size and retrospective study design.

Conclusion

The frequency of additional manual compression and total hemostasis time can be reduced by applying the “US-guided knot advancement technique” when using the SMC device without increased risk of complication.

Declaration of conflicting interests

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